

Fig. 2: Left and right bars displayed on the LCD

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

sons with input signal voltage at analogue input pins

5. Displaying bars and changing their respective heights on the LCD according to the input signals

The program uses special characters for bar-height display on both rows of the LCD using arrays p3, p4, p5, p6 and p7, and arrays L[8] and R[8] for displaying letters L and R on the 16x2 LCD's starting columns of both rows, respectively.

Similarly, array K[8] is used for displaying special characters at the end of both rows, and LEEG[8] is used to display special characters for bar shapes.

LiquidCrystal lcd(12, 11, 5, 4, 3, 2) function is used for LCD configuration.

void setup() function is used to configure the pins of ATmega328 MCU on Arduino UNO board.

Serial.begin(9600) function is used to configure for serial communication with Arduino board at a baud rate of 9600.

lcd.begin(16, 2) function is used

to indicate the size of the LCD.

lcd.createChar(1, p3) function is used to define custom characters for further calling in the program by number only.

lcd.print("VU METER") function is used to display the VU meter as a welcome message.

void loop() function is used to perform the task for infinite

time.

analogRead(inputPin) function is used for reading the analogue-input value.

Simple calculations are performed to get the values for displaying the bar heights as per audio-input signals.

Construction and testing

A USB A-B cable is used to upload the compiled sketch (software) from the PC to the MCU through Arduino IDE.

The operation is as follows:

1. Power on Arduino UNO board using a USB A-B cable. As soon as you power on the LCD, it displays VU meter on the first row and bars on the second row of the LCD.

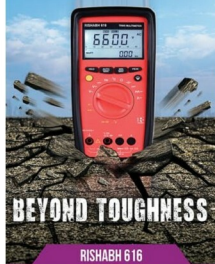
2. 10k potentiometer VR1 can be used to adjust the contrast of the LCD.

3. If we connect the input from a music player to analogue input pins A2 and A4 of Arduino board through two wires, we can easily see the changes in the bars and letters L and R on both rows of the LCD, simultaneously, as shown in Fig. 2. ●



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